MULTI-AREA OSPF

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Purpose

The main objective was to set up multi area OSPF to connect 5 routers in 3 different areas whilst also maintaining connectivity. Running OSPF will also automatically distribute routes throughout the network, meaning all routers should have connectivity with each other. The purpose of this lab is to refamiliarize us with routing protocols and how routing protocol routes are distributed.

Background information on lab concepts

OSPF is designed as an Interior Gateway Protocol. The purpose of which is to be used within an organization's network. Interior gateway protocols work by exchanging routing table information (contains a map of your topology which tells routers how to reach a certain network) with other OSPF configured routers automatically, meaning we do not manually have to set the routes in. The purpose of setting up Multi Area OSPF is to allow for high scalability between different OSPF networks and to maintain redundancy if one or more routers were to suddenly go down. Being an open standard protocol, running OSPF on your routers does not require you to subscribe to a proprietary protocol like EIGRP which is cisco proprietary. Running SPF is also extremely beneficial to which OSPF will automatically find the shortest path to the destination using Link State Advertisements to minimize bandwidth usage. By having all routers in the network run OSPF, the routers will be able to maximize bandwidth and speed. While OSPF is ideal for internal corporate networks and campus networks, however it is not scalable enough for situations such as the intranet for employees. Running multi area OSPFv3 and OSPFv2 will keep IPv4 and IPv6 running at the same time, meaning we can maximize the routing efficiency of IPv6 whilst also maintaining connectivity on devices that are unable to use IPv6 on IPv4. Unlike many routing protocols such as EIGRP and RIP, OSPF does not have limitations in hop count and converges faster in the event of a topology change, and does equal cost load balancing by calculating multiple equal cost paths to maximize the performance of our lines and the redundancy of our lines as if one line goes down, SPF will calculate another path. OSPF's SPF algorithm also calculates loop free paths, meaning information that travels on OSPF links will be safeguarded from loops (lines that might result in a packet traveling infinitely back and forth). Despite all the benefits of Single Area OSPF, it is not

as good at scaling as protocols such as BGP. The largest drawback in OSPF being in comparison to almost every other routing protocol, OSPF is the hardest to manage and troubleshoot. The resources used up by Single Area OSPF are not as much of a drawback but still noteworthy. It takes extra CPU to run SPF and more RAM to store adjacency topology information. resources are resolved. OSPF Link State advertisement information floods the network every 30 minutes which may result in additional traffic taking up bandwidth which may slow down the performance of the network. When running OSPF on large scale networks, the link-state database will greatly expand. This means that even more OSPF link-state advertisements will flood the network bandwidth which might slow down network performance. With a larger topology table, it might overwhelm router's CPU and memory load. OSPFv3 is not compatible with OSPFv2. However, by running Multi Area OSPF, larger routing tables are simplified by keeping link state databases native to the router's current area. Similarly, Link state updates are reduced because updates are only sent to other routers in their area. Therefore, the issues regarding Link state flooding and router

Lab summary

On each router that is running ospf, a network statement is applied which when connected to other OSPF running routers, will automatically distribute the routes to their networks. This means that all routes that are configured with OSPF will automatically be able to communicate and ensure connectivity between these networks. If any routes were to go down (I.e router failure), the redundancy of OSPF's Shortest Path First algorithm will automatically change routes to one that is not down.

Lab Commands

Network Command

The network number of all adjacent links to the current router. These networks are distributed as routes in ospf once adjacency has been established.

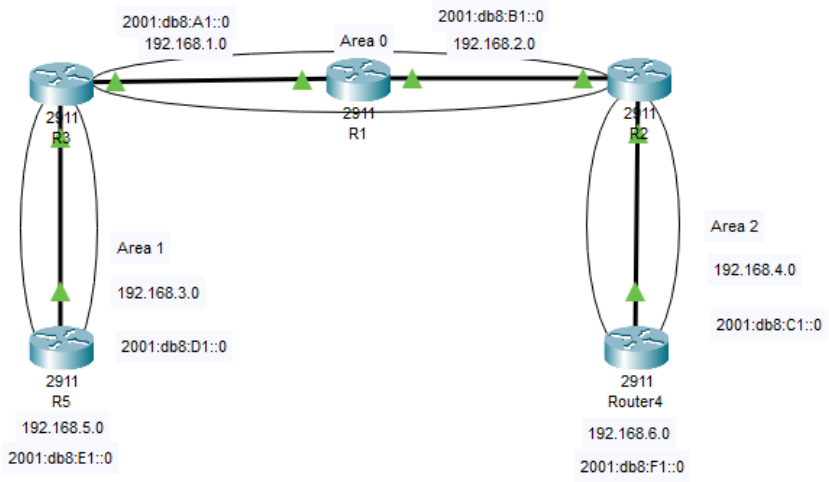
Ip ospf (process id) area (area)

Configures ipv6 routing and distributes ipv6 networks.

Router-id

Configures a router id to influence ospf election processes. Lowest priority is DR.

Network diagrams with IP's



R1

```
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C        192.168.1.0/24 is directly connected,
GigabitEthernet0/0/0

L        192.168.1.1/32 is directly connected,
GigabitEthernet0/0/0

      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks

C        192.168.2.0/24 is directly connected,
GigabitEthernet0/0/1

L        192.168.2.1/32 is directly connected,
GigabitEthernet0/0/1

O IA  192.168.3.0/24 [110/2] via 192.168.1.2, 00:21:19,
GigabitEthernet0/0/0

O IA  192.168.4.0/24 [110/2] via 192.168.2.2, 00:17:02,
GigabitEthernet0/0/1
```

```
      192.168.5.0/32 is subnetted, 1 subnets
O IA     192.168.5.1 [110/3] via 192.168.1.2, 00:21:05,
GigabitEthernet0/0/0
      192.168.6.0/32 is subnetted, 1 subnets
O IA     192.168.6.1 [110/3] via 192.168.2.2, 00:16:27,
GigabitEthernet0/0/1


C   2001:DB8:A1::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:A1::1/128 [0/0]
     via GigabitEthernet0/0, receive
C   2001:DB8:B1::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:B1::1/128 [0/0]
     via GigabitEthernet0/1, receive
OI  2001:DB8:C1::/64 [110/2]
     via FE80::201:43FF:FED9:6B02, GigabitEthernet0/1
OI  2001:DB8:D1::/64 [110/2]
     via FE80::2E0:F9FF:FEB0:3D01, GigabitEthernet0/0
OI  2001:DB8:E1::1/128 [110/2]
     via FE80::2E0:F9FF:FEB0:3D01, GigabitEthernet0/0
OI  2001:DB8:F1::1/128 [110/2]
     via FE80::201:43FF:FED9:6B02, GigabitEthernet0/1
L   FF00::/8 [0/0]
     via Null0, receive
```

R2

```
O     192.168.1.0/24 [110/2] via 192.168.2.1, 00:22:11,
GigabitEthernet0/0/1
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
```

```
C        192.168.2.0/24 is directly connected,
GigabitEthernet0/0/1

L        192.168.2.2/32 is directly connected,
GigabitEthernet0/0/1

O IA  192.168.3.0/24 [110/3] via 192.168.2.1, 00:22:11,
GigabitEthernet0/0/1

      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks

C        192.168.4.0/24 is directly connected,
GigabitEthernet0/0/0

L        192.168.4.1/32 is directly connected,
GigabitEthernet0/0/0

      192.168.5.0/32 is subnetted, 1 subnets

O IA     192.168.5.1 [110/4] via 192.168.2.1, 00:22:11,
GigabitEthernet0/0/1

      192.168.6.0/32 is subnetted, 1 subnets

O        192.168.6.1 [110/2] via 192.168.4.2, 00:21:26,
GigabitEthernet0/0/0

O   2001:DB8:A1::/64 [110/2]

     via FE80::290:CFF:FE3A:3602, GigabitEthernet0/1

C   2001:DB8:B1::/64 [0/0]

     via GigabitEthernet0/1, directly connected

L   2001:DB8:B1::2/128 [0/0]

     via GigabitEthernet0/1, receive

C   2001:DB8:C1::/64 [0/0]

     via GigabitEthernet0/0, directly connected

L   2001:DB8:C1::1/128 [0/0]

     via GigabitEthernet0/0, receive

OI  2001:DB8:D1::/64 [110/3]

     via FE80::290:CFF:FE3A:3602, GigabitEthernet0/1

OI  2001:DB8:E1::1/128 [110/3]

     via FE80::290:CFF:FE3A:3602, GigabitEthernet0/1
```

```
O   2001:DB8:F1::1/128 [110/1]
     via FE80::2E0:A3FF:FEA4:A101, GigabitEthernet0/0
L   FF00::/8 [0/0]
     via Null0, receive
```

R3

```
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected,
GigabitEthernet0/0/0
L        192.168.1.2/32 is directly connected,
GigabitEthernet0/0/0
O     192.168.2.0/24 [110/2] via 192.168.1.1, 00:27:24,
GigabitEthernet0/0/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected,
GigabitEthernet0/0/1
L        192.168.3.1/32 is directly connected,
GigabitEthernet0/0/1
O IA  192.168.4.0/24 [110/3] via 192.168.1.1, 00:23:07,
GigabitEthernet0/0/0
      192.168.5.0/32 is subnetted, 1 subnets
O        192.168.5.1 [110/2] via 192.168.3.2, 00:27:10,
GigabitEthernet0/0/1
      192.168.6.0/32 is subnetted, 1 subnets
O IA     192.168.6.1 [110/4] via 192.168.1.1, 00:22:32,
GigabitEthernet0/0/0
C   2001:DB8:A1::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:A1::2/128 [0/0]
     via GigabitEthernet0/0, receive
O   2001:DB8:B1::/64 [110/2]
     via FE80::290:CFF:FE3A:3601, GigabitEthernet0/0
```

```
OI  2001:DB8:C1::/64 [110/3]
     via FE80::290:CFF:FE3A:3601, GigabitEthernet0/0
C   2001:DB8:D1::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:D1::1/128 [0/0]
     via GigabitEthernet0/1, receive
O   2001:DB8:E1::1/128 [110/1]
     via FE80::2D0:D3FF:FE2A:E901, GigabitEthernet0/1
OI  2001:DB8:F1::1/128 [110/3]
     via FE80::290:CFF:FE3A:3601, GigabitEthernet0/0
L   FF00::/8 [0/0]
     via Null0, receive
```

R4

```
O IA  192.168.1.0/24 [110/3] via 192.168.4.1, 00:25:04,
GigabitEthernet0/0/0
O IA  192.168.2.0/24 [110/2] via 192.168.4.1, 00:25:04,
GigabitEthernet0/0/0
O IA  192.168.3.0/24 [110/4] via 192.168.4.1, 00:25:04,
GigabitEthernet0/0/0
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/24 is directly connected,
GigabitEthernet0/0/0
L        192.168.4.2/32 is directly connected,
GigabitEthernet0/0/0
      192.168.5.0/32 is subnetted, 1 subnets
O IA     192.168.5.1 [110/5] via 192.168.4.1, 00:25:04,
GigabitEthernet0/0/0
      192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.6.0/24 is directly connected, Loopback0
L        192.168.6.1/32 is directly connected, Loopback0
```

```
OI  2001:DB8:A1::/64 [110/3]
     via FE80::201:43FF:FED9:6B01, GigabitEthernet0/0
OI  2001:DB8:B1::/64 [110/2]
     via FE80::201:43FF:FED9:6B01, GigabitEthernet0/0
C   2001:DB8:C1::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:C1::2/128 [0/0]
     via GigabitEthernet0/0, receive
OI  2001:DB8:D1::/64 [110/4]
     via FE80::201:43FF:FED9:6B01, GigabitEthernet0/0
OI  2001:DB8:E1::1/128 [110/4]
     via FE80::201:43FF:FED9:6B01, GigabitEthernet0/0
C   2001:DB8:F1::/64 [0/0]
     via Loopback0, directly connected
L   2001:DB8:F1::1/128 [0/0]
     via Loopback0, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

R5:

```
O IA  192.168.1.0/24 [110/2] via 192.168.3.1, 00:30:35,
GigabitEthernet0/0/0
O IA  192.168.2.0/24 [110/3] via 192.168.3.1, 00:30:35,
GigabitEthernet0/0/0
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/24 is directly connected,
GigabitEthernet0/0/0
L        192.168.3.2/32 is directly connected,
GigabitEthernet0/0/0
O IA  192.168.4.0/24 [110/4] via 192.168.3.1, 00:26:27,
GigabitEthernet0/0/0
```

```
      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.5.0/24 is directly connected, Loopback0
L        192.168.5.1/32 is directly connected, Loopback0
      192.168.6.0/32 is subnetted, 1 subnets
O IA     192.168.6.1 [110/5] via 192.168.3.1, 00:25:52,
GigabitEthernet0/0/0
OI  2001:DB8:A1::/64 [110/2]
     via FE80::2E0:F9FF:FEB0:3D02, GigabitEthernet0/0
OI  2001:DB8:B1::/64 [110/3]
     via FE80::2E0:F9FF:FEB0:3D02, GigabitEthernet0/0
OI  2001:DB8:C1::/64 [110/4]
     via FE80::2E0:F9FF:FEB0:3D02, GigabitEthernet0/0
C   2001:DB8:D1::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:D1::2/128 [0/0]
     via GigabitEthernet0/0, receive
C   2001:DB8:E1::/64 [0/0]
     via Loopback0, directly connected
L   2001:DB8:E1::1/128 [0/0]
     via Loopback0, receive
OI  2001:DB8:F1::1/128 [110/4]
     via FE80::2E0:F9FF:FEB0:3D02, GigabitEthernet0/0
L   FF00::/8 [0/0]
     via Null0, receive
```

Configuration:

R1:

```
Ipv6 unicast-routing

interface GigabitEthernet0/0/0
ip address 192.168.1.1 255.255.255.0
negotiation auto
```

```
ipv6 address 2001:DB8:A1::1/64
ipv6 ospf 1 area 0

interface GigabitEthernet0/0/1
ip address 192.168.2.1 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:B1::1/64
ipv6 ospf 1 area 0

interface Serial0/1/0

interface Serial0/1/1

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

interface Vlan1
no ip address
shutdown

router ospf 1
router-id 1.1.1.1
network 192.168.1.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0

ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0

ipv6 router ospf 1
router-id 1.1.1.1
```

R2

```
Ipv6 unicast-routing
interface GigabitEthernet0/0/0
ip address 192.168.4.1 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:C1::1/64
ipv6 ospf 1 area 2

interface GigabitEthernet0/0/1
ip address 192.168.2.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:B1::2/64
ipv6 ospf 1 area 0

interface GigabitEthernet0/1/0
```

```
no ip address
shutdown
negotiation auto

interface GigabitEthernet0/1/1
no ip address
shutdown
negotiation auto

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

interface Vlan1
no ip address
shutdown

router ospf 10
router-id 2.2.2.2
network 192.168.2.0 0.0.0.255 area 0
network 192.168.4.0 0.0.0.255 area 2

ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0

ipv6 router ospf 1
router-id 2.2.2.2
```

R3

```
Ipv6 unicast-routing
interface GigabitEthernet0/0/0
ip address 192.168.1.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:A1::2/64
ipv6 ospf 1 area 0

interface GigabitEthernet0/0/1
ip address 192.168.3.1 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:D1::1/64
ipv6 ospf 1 area 1

interface Serial0/1/0
no ip address
shutdown

interface Serial0/1/1
```

```
no ip address
shutdown

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

interface Vlan1
no ip address
shutdown

router ospf 1
router-id 3.3.3.3
network 192.168.1.0 0.0.0.255 area 0
network 192.168.3.0 0.0.0.255 area 1
```

R4

```
Ipv6 unicast-routing
interface Loopback0
ip address 192.168.6.1 255.255.255.0
ipv6 address 2001:DB8:F1::1/64
ipv6 ospf 1 area 2

interface GigabitEthernet0/0/0
ip address 192.168.4.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:C1::2/64
ipv6 ospf 1 area 2

interface GigabitEthernet0/0/1
no ip address
shutdown
negotiation auto

interface Serial0/1/0

interface Serial0/1/1

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

interface Vlan1
no ip address
shutdown
```

```
router ospf 10
router-id 4.4.4.4
network 192.168.4.0 0.0.0.255 area 2
network 192.168.6.0 0.0.0.255 area 2

ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0


ipv6 router ospf 1
router-id 4.4.4.4
```

R5

```
Ipv6 unicast-routing
interface Loopback0
ip address 192.168.5.1 255.255.255.0
ipv6 address 2001:DB8:E1::1/64
ipv6 ospf 1 area 1

interface GigabitEthernet0/0/0
ip address 192.168.3.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:D1::2/64
ipv6 ospf 1 area 1

interface GigabitEthernet0/0/1
no ip address
shutdown
negotiation auto

interface Serial0/1/0
no ip address
shutdown

interface Serial0/1/1
no ip address
shutdown

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

router ospf 1
router-id 5.5.5.5
network 192.168.3.0 0.0.0.255 area 1
network 192.168.5.0 0.0.0.255 area 1
```

```
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0

ipv6 router ospf 1
router-id 5.5.5.5
```

Problems

Odd subnetting between all networks.

> Running confusing subnets such as /29 whilst having little organization between these subnets was incredibly confusing until we eventually reset the topology which resulted in a fix. This hit our ipv4 connectivity between routers as our network statements were sometimes wrong.

Cable Failure:

> Some of our ethernet cables were broken wherein one side of our connection between routers was permanently shut down and could not connect with the other side. Due to how rare cable failures are on new cables, troubleshooting our interface disconnects took a while before we came to the correct conclusion.

Unfamiliar commands:

> We used ip ospf 10 area 0 to try and create ipv4 connectivity between routers when the command was unnecessary for ipv4 and instead was to configure ipv6. Unfamiliarity with commands when configuring OSPF sometimes caused us to configure the incorrect commands or use them incorrectly

Bad network topology

> When we set up our previous topology before we restarted, the number of routers in each area was imbalanced significantly because we were unsure how multi-area OSPF would work. It was confusing in which there were three routers in area 2, two routers in area 1, and two routers in area 3. The newest topology has three routers in area 1 and 2 routers in both area 3 and 2. We were more comfortable with this topology because three routers in area 2 allowed us to clean up the uneven side routers and better the overall image of the network.

Unfamiliar with OSPF configuration

The largest issue we faced setting up multi area OSPF is that we were not aware that OSPF required an area 0. due to the nature of OSPF, an area 0 is required because it's the backbone that generates summary routing topology information. When we did not enter an area 0, no routing topology information could be generated which meant that we could not actually ping outside of our area until we restarted.

Conclusion

Whilst using different methods such as static routes may be able to ensure connectivity between our networks, OSPF is one of the most efficient routing protocols to maintain connectivity, redundancy, and speed between our networks. With OSPF's quick network convergence, connectivity between devices will be affected minimally in the event of a topology change such as a failed router. Although running OSPF may seem complicated in how the topology is maintained through routes and link state advertisements, maintaining a complete and accurate topology table and documentation of IP addresses and subnets will ensure OSPF is running optimally and without confusion in the event of future revises

Multi Area OSPF sign off

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Cisco

MASON

Newport High School